

Diseases that affect the pulmonary system can be infections, smoking and inhaled toxic exposures, allergic, auto-immune, occupational, related to treatment (medications/radiation), cancerous, genetic or secondary to other systemic diseases.

Smoking/toxic exposures, Infectious and allergic diseases are the most common in the community. Infections could be viral (COVID-19, influenza, RSV etc), bacterial (Pneumococcus, Hemophilus influenza etc..), tuberculosis, or rarer cases like fungus. 15-20% of the adult population in India are current or prior smokers and the effects of smoking will persist for a life-time. Again, while urban smoking is falling with sustained governmental interventions and campaigns, rural smoking continues to be relatively high due to unregulated beedi and other traditional forms of smoking. Air borne allergens are commonly due to pollens, house dust mites, cockroaches and pets. Food-allergens are relatively uncommon and are due to nuts, eggs, sulphites, sea food and preservatives. The lung can also be affected by systemic organs e.g., the lung is located very close to the heart and can be affected by any disease of the heart, especially the left side of the heart. Auto-immune diseases with arthritis frequently impacts the lung per se or due to its treatment. Liver and gastro-intestinal disease can have involvement of the lung. Lung Cancer causes 10% of all cancer-related deaths in India and is the most common cause of cancer-related death in males in India. Other cancers can spread to the lungs and treatment of cancers can affect the lungs. Other medications like antibiotics, cardiac medications, anti-epileptic drugs and radiotherapy can harm the lung. Most of the respiratory diseases are polygenic i.e., the risk is modulated heavily by the environment. Cystic fibrosis is the most frequent monogenic lung disease and can be as frequent as 1 in 1000 in Westerners. Fortunately, it is rare in the Indian population. The most frequent inherited lung disease is familial interstitial lung disease and 4-10% of all ILDs run in families.

Lungs were the first to be the most affected causing pulmonary failure in those who contracted Covid -19 infection. What leads to such fatal consequences?

A respiratory virus such as Covid-19 attacks the respiratory passage. When the upper respiratory passage is affected by the virus first, it shows up as symptoms of running nose, sore throat, pharyngitis etc. If the virus is mild or the body has strong immunity the virus will be controlled by the body, which happens for most seasonal viral episodes. If a new virus breaks out which the community for which there is no immunity and the virus is potent, then the lower airways can be affected because it

is already present in the upper airway. When the infection enters the lungs, it causes injury and leads to pneumonia. And when the body is still not able to control the infection, it causes alveolar (the part of the lung where oxygen exchange occurs) damage and this is called Acute Respiratory Distress Syndrome (ARDS). Both the lungs are essential to get oxygen into the body and carbon dioxide out of the body every minute of your life and if there is deficit of oxygen supply, your body will cease to perform, leading to fatal consequences.

What are the challenges that you come across while treating a critical patient and are the facilities sufficient to cater to a huge population like in India?

The biggest challenge is equitable distribution of critical care. Unfortunately, critical care is still a nascent specialisation that emerged in our country very recently. Till a decade ago, there was no formal training pathway in the country in critical care and the senior intensivists you see in our country had to train themselves aboard. They have made efforts to train the next generation of intensivists and most of the critical care training in the country is through the efforts of the Indian Society of critical care medicine (ISCCM). Covid has not only heightened the need for critical care facilities such as oxygen therapy, mechanical ventilator and ECMO but also highlighted the need for trained medical personnel to manage it. It is also a challenge because most of the critical care is provided in private hospitals. Critical care is intensive and consequently leads to huge economic burden because of the costs involved. There is need for good critical care services at Government district hospitals upwards and the government is working towards this, especially post-Covid 19. We need to do more and at a much faster pace to be able to cater to our large population. India has become the hub of digital revolution and I believe that the point of interaction between critical care medicine and digital revolution will find all the needed solutions.

The lungs are the purifiers of airy within the body. For the common man what are the essentials that he needs to know about the importance of healthy lungs?

We breathe about 22,000 times a day or about 11000 litres of air every day. You are born with one set of lungs and your life depends on it. Do not smoke or be exposed to smoking passively, eat a balanced diet, do regular exercise and yoga, drink two litres of water and be aware of air quality. The common symptoms in all lung diseases are cough, sputum, blood in sputum, breathlessness, pain in the chest, wheeze and snoring. Cough and sputum are common respiratory disease and patients tend to ignore. Whistling sound indicates infection in

respiratory tract or cough for more than 2 weeks or sputum indicates that there is some irritation in the upper or lower air ways and should be evaluated be a lung specialist. Delays can affect lung function and may reduce survival. Snoring is common in the population but may be sign of sleep apnea, a serious disease which may be associated with uncontrolled hypertension, heart attacks, heart failure and strokes. Snorers should be evaluated for sleep apnea if they have choking during sleep, frequent nocturnal urination, fatigue and sleepiness during daytime.

How advanced is lung transplantation in India, its success ratio and affordability?

For advanced lung disease with respiratory failure, there is no medicine which can reverse the condition. It is an indication that the lungs have failed and needs lung transplantation. The Lungs are a vital organ and progressive lung disease with respiratory failure may have a 2-year survival of <50% for several causes of lung disease. These patients benefit immensely from lung transplantation in the absence of other medical conditions which make transplantation high-risk. The most important condition that increases the risk of lung transplantation is very late presentation and frailty of the patient. Patients able to walk at least 250 meters with oxygen over six minutes are the best stage for lung transplantation; the risk increases with increasing limitation in physical activity. This is because the lungs depend on good chest muscle power to expand and work so that ventilator can be removed after a successful lung transplant. After lung transplantation, 80-85% survive at least one year, 75% survive at least 3-years and 40-50 percent patients survive over 10 years and 30-35 percent survive for 20 years. Lung transplantation saves life and 200 lung transplantation are being performed each year in India each year, including Kauvery Hospitals. It is heartening to note several of our patients doing well and resuming their work and normal life after lung transplantation, with increasing acceptance of this as a treatment modality in our practice.

Transplantation is expensive. Bilateral lung transplants costs Rs 25-35 lakhs, and for a single lung transplant, it is Rs 20-30 lacs. Several states now have support programmes for lung transplantation and several term insurances now cover critical illnesses including lung transplantation. Tamil Nadu is a pioneer and the CM scheme covers critical surgeries like heart and lung transplantation and includes the medication cost for the first year after transplantation. Kauvery Hospital is a leading centre in Chennai which offers this scheme for low-risk patients. Each donated organ is a precious resource and we aim to utilise this with maximum success rate to benefit our patients and the community at large.